

Descriptive Statistics

1. What is the purpose of descriptive statistics?

Ans: Descriptive statistics aim to summarize and describe the main features of a dataset. They provide simple summaries about the sample and the measures. Descriptive statistics are typically used to present quantitative descriptions in a manageable form. This includes measures of central tendency (mean, median, mode), measures of variability (range, variance, standard deviation), and graphical representations (histograms, bar charts, box plots).

2. Can you explain the difference between mean, median, and mode?

- **Mean:** The mean (or average) is the sum of all values in a dataset divided by the number of values. It provides a central value of the dataset but can be influenced by extreme values (outliers).
- **Median:** The median is the middle value of a dataset when the values are arranged in ascending or descending order. If the dataset has an even number of observations, the median is the average of the two middle numbers. The median is not affected by outliers and provides a better central location for skewed distributions.
- **Mode:** The mode is the value that appears most frequently in a dataset. A dataset can have one mode, more than one mode, or no mode at all if no number repeats. The mode is useful for categorical data where we wish to know the most common category.

3. How do you interpret the standard deviation of a dataset?

Ans: The standard deviation measures the amount of variation or dispersion of a set of values. A low standard deviation indicates that the values tend to be close to the mean of the dataset, while a high standard deviation indicates that the values are spread out over a wider range. In other words, standard deviation gives an idea of how much the individual data points deviate from the mean on average.

4. Describe the concept of skewness in statistics.

Ans: Skewness measures the asymmetry of the probability distribution of a real-valued random variable about its mean. It indicates whether the data points are skewed to the left (negative skewness) or to the right (positive skewness).

- **Positive Skewness:** When the tail on the right side of the distribution is longer or fatter than the left side, and the mean is greater than the median.
- **Negative Skewness:** When the tail on the left side of the distribution is longer or fatter than the right side, and the mean is less than the median.

- **Zero Skewness:** Indicates a symmetrical distribution, where the mean and median are equal.

Inferential Statistics

5. **What is the main goal of inferential statistics?** The main goal of inferential statistics is to make inferences about a population based on a sample of data drawn from that population. This involves estimating population parameters, testing hypotheses, making predictions, and drawing conclusions. Inferential statistics allow researchers to generalize findings beyond the immediate data and to determine the reliability and significance of their conclusions.

6. **Explain the difference between a population and a sample.**

- **Population:** A population includes all elements (individuals, items, data points) from a set of interests. It is the entire group that you want to draw conclusions about.
- **Sample:** A sample is a subset of the population, selected for analysis. Samples are used in statistical analysis because it is often impractical or impossible to study the entire population. The goal is to use the sample data to infer information about the population.

7. **What is a confidence interval, and how is it useful in inferential statistics?**

Ans: A confidence interval is a range of values, derived from sample data, that is likely to contain the value of an unknown population parameter. The interval has an associated confidence level that quantifies the level of confidence that the parameter lies within the interval. For example, a 95% confidence interval means that if the same population is sampled multiple times and interval estimates are made, 95% of the intervals would be expected to contain the population parameter. Confidence intervals are useful because they provide an estimated range of values that is likely to include the population parameter, giving a sense of the precision and reliability of the estimate.

8. **Define p-value.**

Ans: The p-value is the probability of obtaining test results at least as extreme as the observed results, assuming that the null hypothesis is true. It is a measure of the evidence against the null hypothesis. A low p-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, so it is rejected. Conversely, a high p-value (> 0.05) indicates weak evidence against the null hypothesis, so it is not rejected. The p-value helps in determining the significance of the results in hypothesis testing.